

leptons 12, 21, **123**, 124–25
 levitation, magnetic **94**
 light
 absorption 86
 charge-coupled devices **93**
 coherent 86, **87**
 electromagnetic radiation **14**, **22**
 lasers **87**
 light-emitting diodes (LEDs) 90, **92**
 photoelectric effect **32–33**
 photosynthesis **148–49**
 speed of 15, 74, 80
 stimulated emission 86
 light energy 148
 light quanta 28, 30, 32
 loop quantum gravity **144–45**

M

M-theory **143**
 Maglev transport systems 94
 magnetic fields
 electromagnetic radiation 14
 electron microscopes 100, 101
 MRI 98–99
 quantum navigation **152–53**
 SQUIDS 96–97
 subatomic particles 67
 superconductors **94**
 magnetic flux 96
 magnetic moments **67**, 99
 magnetic resonance
 imaging 94, **98–99**
 magneto-optical traps 82
 magnets, superconductors **94**
 many worlds interpretation **54**
 mass, particles 127

matter
 and antimatter 128–29
 atoms 10
 electromagnetic radiation 14, 15
 and light 25
 unusual states of 65
 matter particles 14, 20, 21, 68, 122, 124–25, 128
 matter waves 56, 100
 measurement problem **46–47**
 mesons 12, 13, 135
 metals, photoelectric effect **32–33**
 microwaves 14, 88
 migrations, seasonal 152–53
 momentum 41, 42–43
 MRI scanners see magnetic resonance imaging
 multidimensional space 143
 multiple states 38
 multiple universes 54, 55
 muon neutrinos 12, 123, 125
 muons 12, 124

N

nanoscale 102–103
 neutrinos **123**, 124–25
 neutron stars 117
 neutrons 8, 10, 12, 13, 20, 21, 70, 112–13, 122
 nuclear fission 114–15
 nuclear fusion 116
 Newton, Isaac 35, 45
 nuclear fission 111, **114–15**
 nuclear fusion 71, 111, **116–17**
 nuclear physics **111–19**
 nuclei, atomic 20, 111
 nuclear fission **114–15**
 nuclear fusion **116–17**
 size 8, 9
 unstable 21, 65, 112–13
 nucleons 8

O

objective collapse hypothesis **59**
 observable universes 55
 observed/unobserved events 52–53, 59, 61, 63
 olfaction **154**
 open vibrating strings 142
 optical amplification 87
 oscillations 14, 88–9

P

p-orbitals 31
 parallel universes **54**, 55
 particle accelerators **121**, 130
 particle detectors 18, **120**, 121
 particle physics **119–35**
 particle-wave duality **18–19**
 Pauli exclusion principle **69**, 79
 photodiodes 102
 photoelectric effect **32–33**
 photons 13, 18, 22, 25, 46–47, 108, 113, 124, 126, 127, 152
 charge-coupled devices 93
 color and energy 30
 LEDs 92
 photoelectric effect **32–33**
 photosynthesis **148–49**
 Planck's constant **15**
 polarization 106
 quantum electrodynamics **134**
 stimulated emission **86**
 photosynthesis **148–49**
 pilot wave theory **56–57**
 pions 13
 pixels 93
 Planck, Max 28
 Planck length 9, **140–41**
 Planck scale 9, 140–41, 144